



Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
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Practical Exercise no. 10

Theme Support Vector Machines

Exercise Support Vector Machines (SVMs) are a type of supervised learning model that can be used for classification or regression problems, or to detect outliers. The technique creates hyperplanes that separate the cases under study into classes. The aim is to identify the hyperplane with the greatest possible margin, i.e. the maximum distance between the data points of the two classes. SVMs can solve both linear and nonlinear problems, producing accurate models with less computational effort.

Tasks This practical exercise includes the following tasks:

T1. Follow and implement the steps presented in the practical notebooks to apply the Support Vector Classifier to the *GenAudio* dataset. Try to optimise the model by varying additional hyperparameters. *GridSearchCV* can be used to help select the optimal hyperparameter configuration. What is the optimal SVC configuration? How well did the model perform when tested on an audio file?

T2. Follow the steps in the practical notebooks to design and optimise an SVM model using the datasets from the practical group work.